

Question

The X-ray diffraction data (2θ) of a certain substance measured by Xiao Ming are as follows:

32.2° , 37.3° , 53.9° , 64.2° , 67.3°

It is known that the test was conducted using Cu - K α radiation. Now, there are several statements:

1. Assuming the relative molecular mass of the substance is 100, its crystal density is approximately $5.97 \text{ g} \cdot \text{cm}^{-3}$
2. Among the above diffraction peaks, there are second-order or higher-order diffraction peaks
3. To capture more diffraction peaks within $5 - 80^\circ$, the Cu target of the instrument needs to be replaced with Mo
4. To capture more diffraction peaks within $5 - 80^\circ$, the Cu target of the instrument needs to be replaced with Cr
5. Without changing the test conditions, the next diffraction peak theoretically appearing at a higher angle would be from the (331) plane

Among the above statements, the correct ones are

A. All statements are incorrect

B. 1

C. 2

D. 3

E. 4

F. 5

G. 1,2

H. 1,3

I. 1,4

J. 1,5

K. 2,3

L. 2,4

M. 2,5

N. 3,5

O. 4,5

P. 1, 2, 3

Q. 1, 2, 4

R. 1, 2, 5

S. 1,3,5

T. 1,4,5

U. 2,3,5

V. 2,4,5

W. 1, 2, 3, 5

X. 1,2,4,5

Answer

Correct Answer: **P**

Detailed Explanation

By default, it is known that the wavelength of Cu-K α radiation is 154.1pm.

First, according to the Bragg diffraction equation:

$$2d \sin(\theta) = n\lambda$$

the interplanar spacings corresponding to the given angles can be calculated as:

278 pm, 241 pm, 170 pm, 145 pm, 139 pm

Since

$$d_{(hkl)} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

by squaring the above interplanar spacings and taking their ratios, we obtain a set of approximate integer ratios, which correspond to the ratios of $h^2 + k^2 + l^2$:

3 : 4 : 8 : 11 : 12

CHECKPOINT

1 PTS

The ratios of $h^2 + k^2 + l^2$ are approximately 3 : 4 : 8 : 11 : 12

This matches the extinction rules for a face-centered cubic (FCC) structure, indicating that the material has an FCC lattice.

CHECKPOINT

2 PTS

The material has a face-centered cubic structure

The five diffraction peaks correspond to the (111), (200), (220), (311), and (222) crystal planes, respectively.

CHECKPOINT

1 PTS

The five diffraction peaks correspond to the (111), (200), (220), (311), and (222) crystal planes

For Statement 1:

Using the formula: $d_{(hkl)} = \frac{a}{\sqrt{h^2+k^2+l^2}}$

we obtain the unit cell edge length $a = 481 \text{ pm}$.

From the formula $\rho = \frac{ZM}{VN_A}$, the density is calculated as $5.97 \text{ g} \cdot \text{cm}^{-3}$, which is correct.

CHECKPOINT

1 PTS

The density is $5.97 \text{ g} \cdot \text{cm}^{-3}$

CHECKPOINT

0.5 PTS

Statement 1 is correct

For Statement 2:

The diffraction peak of the (222) plane can be considered as the second-order diffraction of the (111) plane, which is correct.

CHECKPOINT

0.5 PTS

The diffraction of the (222) plane can be regarded as the second-order diffraction of the (111) plane

CHECKPOINT

0.5 PTS

Statement 2 is correct

For Statements 3 and 4:

According to the formula: $2d \sin(\theta) = n\lambda$

For the same material where d remains unchanged, a smaller λ corresponds to a smaller θ .

CHECKPOINT

1 PTS

For the same material where d remains unchanged, a smaller λ corresponds to a smaller θ

The wavelength of Cu target is 1.54 Å; Mo target is 0.71 Å; Cr target is 2.29 Å.

Thus, Statement 3 is correct, and Statement 4 is incorrect.

CHECKPOINT

1 PTS

Statement 3 is correct, Statement 4 is incorrect

For Statement 5:

Theoretically, the next crystal plane to exhibit diffraction should be the (400) plane with $h^2 + k^2 + l^2 = 16$, not the (331) plane with $h^2 + k^2 + l^2 = 19$. Therefore, Statement 5 is incorrect.

CHECKPOINT

1 PTS

The next diffraction plane should be the (400) plane

CHECKPOINT

0.5 PTS

Statement 5 is incorrect

Choose P.